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# =====
# Facial Recognition Pipeline using OpenCV
# Google Colab Compatible
# =====

import cv2
import matplotlib.pyplot as plt
from google.colab import files

# -----
# Step 1: Image Acquisition
# -----
print("Upload a face image")
uploaded = files.upload()

image_path = list(uploaded.keys())[0]

# Read the uploaded image
image = cv2.imread(image_path)

if image is None:
    print("Image not found!")
    exit()

# Convert BGR to RGB for display
rgb_image = cv2.cvtColor(image, cv2.COLOR_BGR2RGB)

plt.figure(figsize=(5,5))
plt.imshow(rgb_image)
plt.title("Step 1 : Acquired Image")
plt.axis("off")
plt.show()

# -----
# Step 2: Low-Level Computer Vision
# Image Preprocessing
# -----

# Convert to grayscale
gray = cv2.cvtColor(image, cv2.COLOR_BGR2GRAY)

# Reduce noise
gray = cv2.GaussianBlur(gray, (5,5), 0)

# Improve contrast
gray = cv2.equalizeHist(gray)

plt.figure(figsize=(5,5))
plt.imshow(gray, cmap="gray")
```

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plt.title("Step 2 : Preprocessed Image")
plt.axis("off")
plt.show()

# -----
# Step 3: Mid-Level Computer Vision
# Face Detection
# -----

face_detector = cv2.CascadeClassifier(
    cv2.data.haarcascades +
    "haarcascade_frontalface_default.xml"
)

faces = face_detector.detectMultiScale(
    gray,
    scaleFactor=1.1,
    minNeighbors=5,
    minSize=(30,30)
)

output = rgb_image.copy()

for (x,y,w,h) in faces:
    cv2.rectangle(output,
                   (x,y),
                   (x+w,y+h),
                   (0,255,0),
                   3)

plt.figure(figsize=(6,6))
plt.imshow(output)
plt.title("Step 3 : Face Detection")
plt.axis("off")
plt.show()

# -----
# Step 4: High-Level Computer Vision
# Decision Making
# -----

print("===== FINAL DECISION =====")

if len(faces) > 0:
    print("Face Detected")
    print("Identity verification can now be performed.")
else:
    print("No Face Detected")

print("Number of Faces Detected:", len(faces))
print("=====")
```

Upload a face image

OKV2u4evL-_j60T3aLQcTXkNsM7NzokTyjjJFG5Sbs7FYY7_BlkBT7MyZhtRm5IE_RLk_x_D_

OKV2u4evL-

_j60T3aLQcTXkNsM7NzokTyjjJFG5Sbs7FYY7_BlkBT7MyZhtRm5IE_RLk_x_D_qXEzR1HK

cC4Z-WSzPJWFwNXNF4MLxoy78PNB8792GlgKYC5fD0aUtpw3g.jpg(image/jpeg) - 28985

bytes, last modified: n/a - 100% done

Saving OKV2u4evL-_j60T3aLQcTXkNsM7NzokTyjjJFG5Sbs7FYY7_BlkBT7MyZhtRm5IE_RLk_x

Step 1 : Acquired Image

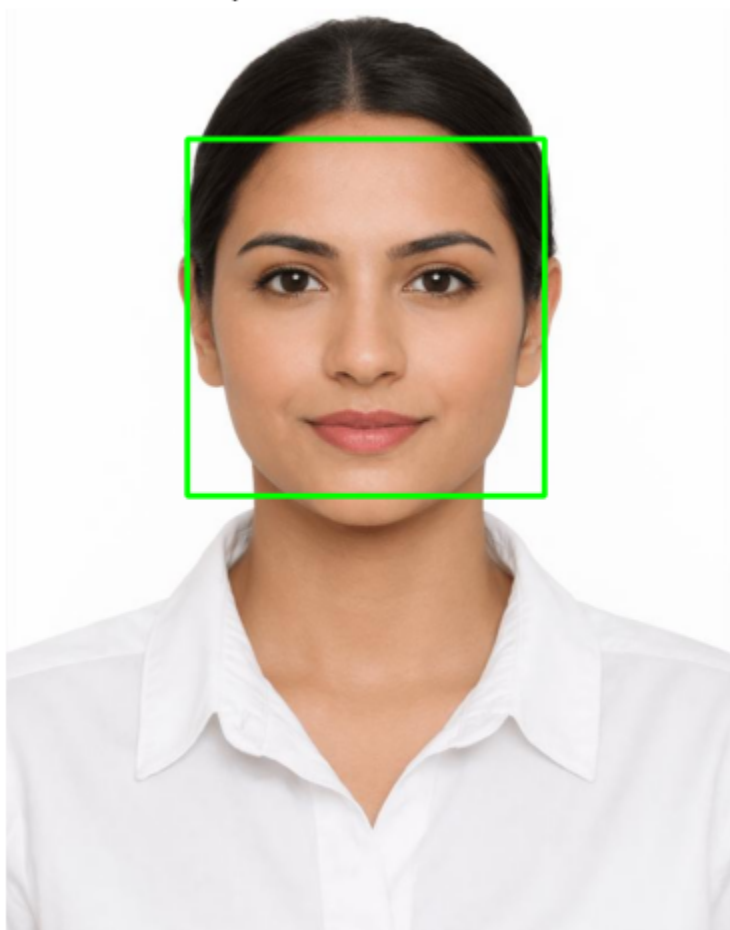


Step 2 : Preprocessed Image





Step 3 : Face Detection



```
===== FINAL DECISION =====  
Face Detected  
Identity verification can now be performed.  
Number of Faces Detected: 1  
=====
```